

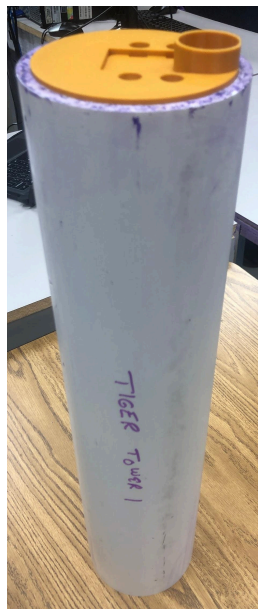
Tiger Tower I Logbook

Written By Joshua Salu

Foreword

This is a public copy of the logbook I made for the vertical profiling float that is to participate in the Houston MATE 2026 competition under the Kerr Engineering Club, also known as Kerr Robo Incorporated. This logbook will often reference links and files that may not be accessible; however, the intent of this logbook is to give insight into the construction of the float without external resources.

Additionally, while there was initially supposed to be a design team, due to student-related circumstances as well as a lack of time, all of the work related to the vertical profiling float was delegated to me, while all other team members were to work on the ROV. While I was unable to finish the float due to the urgency of the ROV, I have made this logbook public so that future members of the team can learn from my efforts.



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Beginning Task #4

Today, the meeting was dedicated to coming up with a design for the vertical profiling float in task #4 of the MATE ROV competition.



Example of a Vertical Profiling Float

Task #4 involves constructing a **vertical profiling float** separate from the ROV to submerge in water and complete vertical profiles autonomously. The reason why I decided to prioritize this first is that this component is separate from all other tasks. To elaborate, tasks 1-3 are only possible with the ROV. Additionally, the ROV itself is being developed by Abdul right now, meaning very few tasks can be progressed. Therefore, it would be best to develop the float simultaneously.



One of the first things decided was that **I would use a PVC pipe as the chassis**. PVC is sturdy, reliable, cheap, and readily available. There are multiple ways to accomplish the construction of a vertical profiling float, and one of the key decisions in this process is the choice of our buoyancy engine. However, this choice, and the rest of this process, is complex and will require additional research to make some drafts.

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After conducting extensive research, I have split up the vertical profiling float into 3 parts: the chassis, the electronics compartment, and the buoyancy engine.

Chassis

The chassis is mostly simple in concept: it protects all internal components of the float. However, where it gets complex is the required holes, sealing, and pressure relief.



Note: This is not the real sensor being used.

The chassis requires at least one hole for the pressure sensor, which will be required to record data during each vertical profile. An additional hole will also need to be made for a switch, so that it can be deployed by hand. The diameter and location of this hole are currently unknown, although it will likely be within the electronics compartment. The exact means of sealing the chassis is currently unknown. Any necessary holes will be sealed with silicon or epoxy. **However, the float itself requires some form of pressure relief. There are two ways to accomplish this: corks or o-rings.** Corks (or rubber stoppers) can be friction-fit into a hole to prevent fluids from coming in while allowing for pressure relief. O-rings are rubber gaskets that fulfill a similar function, but only when fit to the material. (In other words, the o-ring must be wrapped around something so it can seal the compartment.) Corks are easier to remove and modify, but take up additional space. O-rings take up less space but cannot have their circumference manipulated by any means. **As of now, the current diameter of the chassis is 10 cm, and its height is around 56 cm.**

Electronics Compartment

The electronics compartment is meant to power the float so that it can record vertical profiles automatically. The key components of this compartment are the power source, the fuse, the motor, the pressure sensor, the breadboard, and the microcontroller. Currently, I am planning to use 4 AA NiMH batteries to power the float. This will provide us with a voltage of around 4.8V. **NiMH (Nickel Metal Hydride) batteries are one of two allowed types of batteries for providing power.** NiMH batteries can also be recharged. As per MATE's safety regulations (ELEC-NRD-003), **I must use a 2 amp fuse** (if I am using NiMH batteries), **and it must be within 5 cm of the power source.** These batteries will also power a 5V-12V motor, but the details of the motor itself are currently unknown. The pressure sensor will also require power of its own, and its details are also unknown. The breadboard itself will likely be a typical half-size perma-proto breadboard. **The microcontroller itself will be a Raspberry Pi Pico W.** This microcontroller was chosen due

to its compact size, low price, and innate wireless capabilities. Should the wireless capabilities be insufficient, a radio module can be connected.

Buoyancy Engine

The buoyancy engine is the most important part of the float, as it decides the design of the majority of the float, as well as how it will manipulate the float to conduct vertical profiles. For context, all objects on Earth are subject to gravity, which is mass times the acceleration from gravity (9.81 m/s^2). All objects submerged in water experience a force of buoyancy opposite to gravity, which is the product of fluid density, gravity, and the volume of the object. For an object to float, the buoyant force must exceed the gravitational force. The opposite is true for an object to sink. An object will remain in equilibrium if the gravitational force and buoyancy are the same. In other words, if those conditions are met, the object will neither sink nor float and remain still, which is vital to taking vertical profiles at any given depth.

There are two main types of buoyancy engines: water-based engines and air-based engines. Both manipulate the force of buoyancy within water to float, sink, or remain neutral. **Water-based engines** use either a **large syringe or a hydraulic cylinder** to intake water, increasing the mass of the float to strengthen the force of gravity. **Air-based engines** use a **bladder and an air tank** (or some other fluid) to expand the bladder, increasing the volume of the float to strengthen the force of buoyancy. It should also be noted that water is many times denser than air. To be specific, water has a density of $1,000 \text{ kg/m}^3$, while air has a density of about 1.225 kg/m^3 at sea level. (Density can vary based on conditions, like temperature). Given our limitations, **it is likely best to design a water-based buoyancy engine using syringes.**

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We have made the decision to use a water-based buoyancy engine for the vertical profiling float. Because I am using syringes, it is guaranteed that I will use a motor to construct **a mechanism that converts rotational motion into translational motion.** To elaborate: as the motor revolves, an object connected to that actuator will move forward and backward.

Syringe Choices

There are two potential designs to consider: one using smaller syringes and one using a custom, larger syringe.



The former involves **using readily available syringes to intake water**. They will take up less space and require less force to push down. In exchange, the syringes cannot intake as much volume of water, which could be insufficient for functionality in the worst-case scenario. In the average case scenario, this could lead to a slow descent, which is not inherently negative but should be accounted for.

Currently, a syringe holder has been designed to fit **3 60 mL syringes**, while also functioning as a cork to do away with the need for adhesives. (This could change in the future.)

A larger custom syringe will be harder to implement, as it will require a friction-fit disk, as well as more power from the motor that could exceed our given limit. It will also be larger, requiring more space in the chassis. In exchange, a larger syringe can ensure that the float can sink or float, though finding neutral buoyancy may be harder as a result.

Currently, I will go with the first design *unless* it is found that the changes in mass provided by the smaller syringes are insufficient.

Linear Motion Mechanisms

The mechanism itself can also have two (or more) designs. One option is to use a **linear actuator** using a screw mechanism. However, the implementation of this design is complex and must be researched. Additionally, a linear actuator must be designed rather than bought. A bought linear actuator will increase the cost by a great amount, while also potentially being unfit for our float. Creating a linear actuator will give us more agency in the design while also keeping costs low.



Therefore, it may be simpler to go with a slider-crank linkage (or other similar mechanisms like the sun-planet gear or scotch yoke). All three mechanisms use an apparatus with a motor attached to the center and a rod being held to prevent vertical

movement, resulting in the rod oscillating horizontally. (Implementation of the linear actuator is not limited to these options; these are merely suggestions at the time of writing. All given flaws of these mechanisms are not fully realized as of yet.)

In both cases, the linear actuator will be attached to a piston that will connect to the syringe(s).

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For now, I've shifted my focus to the construction of the electronics compartment. As mentioned earlier, the key components of this compartment are the power source, fuse, motor, pressure sensor, breadboard, and microcontroller. The switch will also be considered a component of the electronics compartment. This switch can be implemented in two ways.

Switch Designs

Lever Switch w/ Cap

Pros	Cons
• PVC structures already exist that can <u>make implementation easy</u> • <u>Easy to tell if the float is on or off</u>	• <u>Cannot be toggled at a moment's notice</u> • <u>Is not sealed at all points in time</u>; improper sealing can cause extreme damage w/o safeguards • Would <u>likely require a protrusion</u>; depressing the material could worsen the previous con

Push-Button Switch w/ Membrane

Pros	Cons
• <u>Faster toggling</u> • <u>Always sealed</u>	• <u>Harder implementation time</u>; the design would have to consider many factors on the outside, compared to a lever switch • <u>May require a visual indicator</u> to tell if the float is powered or not

For now, I will go with the lever switch. Ideally, the lever will be small, such that it can be toggled with one finger. This way, I could end up using a membrane instead of a cap (if possible).

To allocate space for this compartment, the breadboard and microcontroller will be given a container (dubbed the electronics hub) so that all wiring is connected and made easily accessible. Additionally, the power source will be kept separate from this hub to

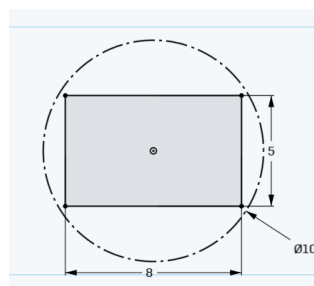
accommodate the removal of batteries. The fuse, consequently, will also have to be kept outside of the hub to comply with regulations. To elaborate, the fuse **must be within 5 cm of the power source.** This also means that the switch will need to be kept in close proximity to the power source. The motor and pressure sensor will also be kept separate from the hub, but their wiring will connect to the hub to be controlled by the microcontroller. Finally, it was found that powering the motor and pressure sensor using the 3.3V out pin of the Raspberry Pi Pico W is inadvisable. **Therefore, I will supply power to these devices separately,** which means that I will be using 9-12V instead of the planned 4.8V. This may also involve the use of a step-down transformer within the circuit. With this in mind, I have begun to design potential designs and prototypes for the electronics hub.

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Custom designs for the electronics hub have been drafted. The hub itself can be made of currently existing objects, like jars or PVC pipes. They are easy to seal and drill holes in, but their size and design cannot be customized easily. *While this option isn't ruled out* (by virtue of convenience), I decided to create designs on a platform called **Onshape.**

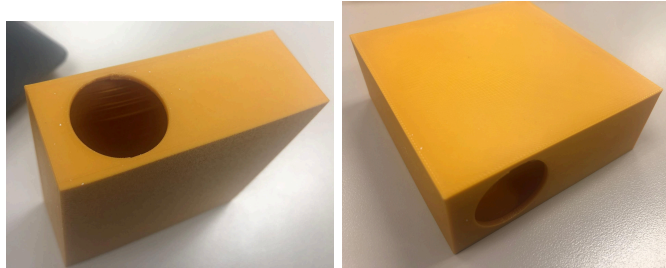
Onshape is a free CAD system that uses cloud architecture. We've decided to use this program because it can be accessed on any computer, and members can collaborate on the same design.

To make a competent design, I would decide on using 3D printed prototypes for precise measurements and features. Additionally, 3D prints are not perfect, and may have defects that may or may not be able to be fixed. Any imperfections or oddities may also affect how I approach the remaining tasks. Finally, these designs will also help us evaluate the use of PLA and other 3D printing filaments to make the shell, as well as other components. Therefore, I have come up with designs to observe and test various characteristics that can be utilized in the final design. Using Onshape, we've been able to fully design 2 prototypes for the hub.



As a precedent, **no hub design should exceed 8 cm in width or 5 cm in length.** This is to respect the 10 cm diameter.

1st E. Hub Prototype

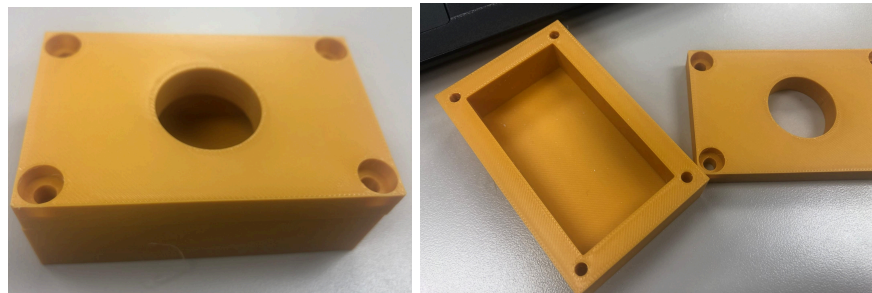


(Note: For testing, a shell with one hole will be used.)

This prototype uses thin walls and largely relies on a cork to provide pressure relief. However, upon printing, I noticed that side-printed circles are prone to imperfections. Additionally, the pressure relief hole itself is comparably shallow, only 0.25 cm in depth. This could affect a cork's ability to seal the compartment. The pressure relief hole itself has a slight defect, as well. This could be due to being printed on the side, instead of at the top or bottom.

Additionally, this design cannot currently be opened, though this limitation could be mitigated by accommodating adjustments only to the microcontroller, sensor, and battery.

2nd E. Hub Prototype



This prototype uses a thicker design that is separated into two parts to be combined with screws. This design allows for revision of components, although using screws is not the only way. Additionally, the thicker design does leave less space for electronics, though this can be mitigated by separating the battery and pressure sensor from the shell. Doing this, however, has its own complex requirements to ensure modularity.

Finally, the actual fabrication showed that the bases were not made flat, which could heavily reduce the effectiveness of these screws.

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Today, two tests have been carried out on both prototypes: a durability test and a sealing test.

Durability Test

Rationale

The durability test was conducted to ensure the compartment can withstand minor forces within the float. This is to make sure that this compartment isn't destroyed before pressure is relieved, and that it can retain its function should the float be damaged (or rocked) from the outside. The durability of the compartment shell will mostly rely on thickness, so failing this test would require a revision of the shell's thickness. In other words, this test is meant to optimize the thickness of the shell's walls. If a pre-existing container is used instead, then this test can be used to evaluate an acceptable degree of durability for said container.

Procedure

The shell will be subjected to 5 lbs of force for 15 seconds over the course of three trials. If there is no sign of damage (bending, cracking), then the shell passes the test.

Data

	Trial 1 (5s)	Trial 2 (10s)	Trial 3 (15s)
Prototype 1	Pass	Pass	Pass
Prototype 2	Pass	Pass	Pass

Sealing Test

Rationale

Instead of another O-ring (that may need to be custom-fit within the file itself), a cork can be used to seal the shell. The cork will be sufficient for pressure relief, but it may also need protection from water, should it enter the float. If the seal fails, then any water will compromise the entire float. This test can also help evaluate if the location or type of hole needs to be changed, or if another method should be used entirely. Should both designs

fail this test, then a pre-existing container will likely be used since most already provide competent sealing.

Procedure

The shell will be sealed with a cork. The shell will be exposed to light, running water from above for 5 seconds, with the cork side facing up. Then, the shell will be unsealed. If the shell has no water inside, then the shell passes the test. In each trial, the cork should be adjusted to ensure maximum sealing.

Data

	Trial 1	Trial 2	Trial 3
Prototype 1	Fail	Fail	Fail
Prototype 2	Fail	Fail	Fail

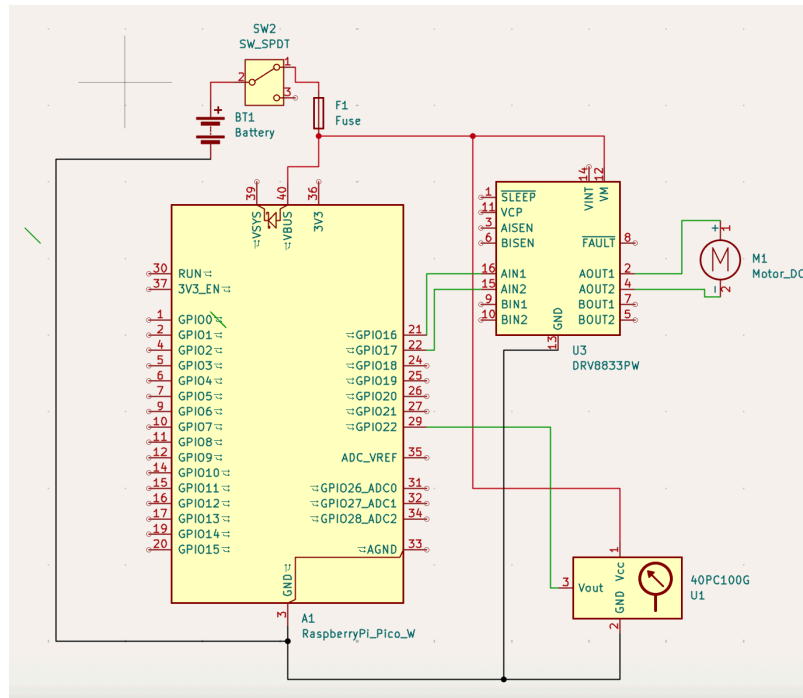
Conclusion

The results of the durability test showed that Prototype 1's thickness was sufficient in withstanding minor forces beyond 15 seconds. Prototype 2, naturally, was also sufficient, but the results of Prototype 1 prove that thick walls are not necessary to provide resistance against forces.

The results of the sealing test showed that the usage of corks may not be as viable for protection from water. Despite this, corks can still be used, although this may require more precise fitting (and even attachments). Alternatively, water protection can be sacrificed entirely, and the cork can be oriented near the bottom to use gravity. Ideally, a solution can be found that provides water protection and accounts for any known defects of the prototypes, while also being non-permanent to allow for modifications to the electronics.

In any case, pre-existing containers will be used from now on. However, these findings are useful for the evaluation of the hub's requirements and the potential benefits and flaws of our 3D prints.

Concerning the vertical profiling float's buoyancy engine, it has been decided that a **slider-crank mechanism** will be used as a linear motion mechanism. Out of the [named mechanisms](#) (save for the linear actuator), the slider-crank mechanism can handle the most load. However, the motor will likely need to be oriented sideways, meaning that additional mechanisms must be put in place to align the rod in the middle for the piston. This could make fabrication more difficult than expected. However, this does give us the opportunity to create gears to apply a stronger linear force, which is something that would've needed to be done for both designs anyway. Additionally, a week ago, an initial circuit design was created.



Certain components were accounted for, such as the separation of the motor and pressure sensor, the addition of a motor controller (DRV833), and the use of a Raspberry Pi Pico W. Further research revealed that the **Raspberry Pi Pico W is not suitable for underwater communications**. While it is a known limitation that frequencies can be absorbed by water, this can be mitigated by restricting communication to only occur when the float is not fully submerged. Additionally, placing radios in a plastic compartment is safe, since signals can pass through. A longer-range alternative would be the **Adafruit Feather RP2040 with an RFM95 LoRa Radio**. This uses the same RP2040 chip that a Raspberry Pi Pico W uses, while boasting an alternative radio for long-range communication. However, this is more expensive than the Pico (by about 4 times). Additionally, I can connect a separate radio module and antenna. The choices of using a different microcontroller, using a separate module, or keeping the Raspberry Pi Pico W are yet to be explored as of writing.

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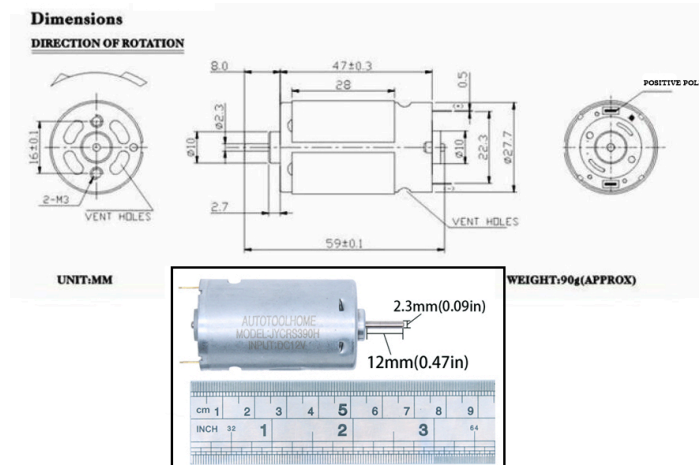
Additional Float Update

The decision to use a slider-crank mechanism has been rescinded. Research showed that while it *can* be used in syringes, it lacks the precision required to manage the buoyancy of the float, which is extremely vital to this task. Additionally, two linear actuator designs have been found that are less complex to design than previously anticipated. In any case, **a linear actuator will be used** for the buoyancy engine to ensure precision when controlling buoyancy.

The construction of the linear actuator will require the use of a motor, bearings, a threaded bolt, and a driver nut. The actuator transforms rotational motion into linear displacement by having the motor rotate the bolt. The bolt itself will not experience linear displacement due to being held down; instead, an arm will extend via the driver nut. The extension of this arm is slow, but in exchange, the torque provided by the motor will be converted into a high amount of force. The arm will have a piston so that this force can be used to push and pull the syringe. It is important that the force is sufficient to push down a syringe filled with water, especially if it has a relatively large volume. This is because water is many times denser than air, requiring more work to push the water from the syringe.

Buoyancy Engine Motor

The motor I will be using for the float is the AUTOTOOLHOME 6V-12V DC motor.



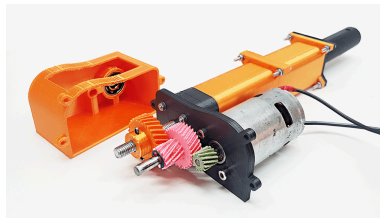
Dimension	Performance		
	Voltage	Deprating Range	V/DC
			6V-18V
	AT NO LOAD	Voltage	V/DC
			12V
		Speed	r/min
			12000
		Current	A
			0.24
	AT MAXIMUM EFFICIENCY	Efficiency	%
			no
		Speed	r/min
			10386
		Current	A
			1.55
		Torque	g.cm
			96
		Output	W
			10.23
	AT MAX POWER	Output	W
			22
		Speed	r/min
			6000
		Current	A
			5.12
		Torque	g.cm
	AT STALL		357
		Torque	g.cm
			714
		Current	A
			10

Also, starting today, the name of the vertical profiling float will be the Gargoyle I.

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Linear Actuator Design

Today, I gathered materials to begin the design process of the linear actuator. We found an M7 bolt with the corresponding nuts, as well as some bearings. The first task I decided on was to begin designing a gearbox.



Design by Retsetman

Instead of mounting the bolt to the motor directly, it would be superior to use a **gearbox**. A gearbox allows us to convert rotational speed into torque, while also allowing for flexibility with the orientation of the motor.

The **gear ratio** is critical to the force the linear actuator can deliver. Gear ratios dictate the amount of torque exerted in relation to the rotational speed. The leftmost number is the number of revolutions made by the **driver gear**, which is the gear being rotated by the motor, while the rightmost number is the number of revolutions made by the **output gear**. For example, a 1:2 gear ratio means the output gear rotates 2 times for every one rotation from the driver gear. This means that the speed is doubled, but the torque exerted is halved. Conversely, a 2:1 gear ratio means that the output gear rotates once for every 2 rotations from the driver gear. This means that the torque exerted is doubled, while the speed is halved.

For our application, the exerted torque (and thus force) is favored over speed. The decrease in speed alone actually benefits the float, since this allows for more precision in achieving neutral buoyancy. However, this does not mean an extremely high gear ratio should be used. As the gear ratio increases, friction will increase, straining the motor, consuming more power, and causing the movement to become inconsistent, provided it doesn't stop entirely.

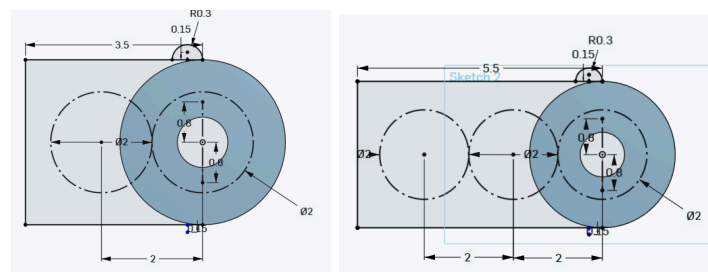
Many electric linear actuators have gear ratios ranging from 10:1 to 100:1. While the gears themselves have not been finalized, I have 3D printed gears of a 2 cm diameter with 27 teeth. **Gear teeth** are essential for evaluating the gear ratio of a gearbox. For example, if the driver gear has 9 teeth and the output gear has 27 teeth, the gear ratio is 3:1. I currently plan to use a 10:1 gear ratio at maximum.

Lubricant is also a notable component of linear actuators. Most linear actuators possess lubricant in both the arm and the gearbox to ensure smooth motion. Should lubrication be necessary, food-grade silicone spray/grease can be used; however, I have not yet determined whether lubrication is required at this time. This is because it is highly likely that I will be using 3D-printed gears rather than gears carved from wood or made from metal.

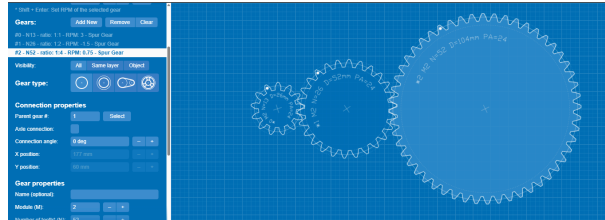
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Actuator Gearbox Design

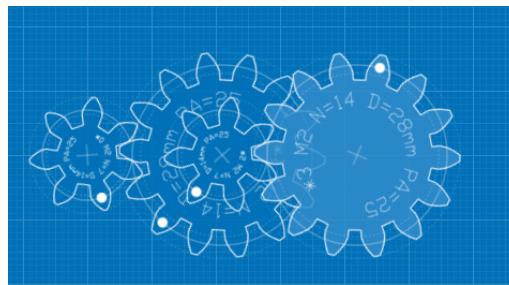
In Onshape, a few designs were drafted in an attempt to experiment with the possible combinations of gears I could make. I ended up with two main designs, one with two gears and one with three gears.



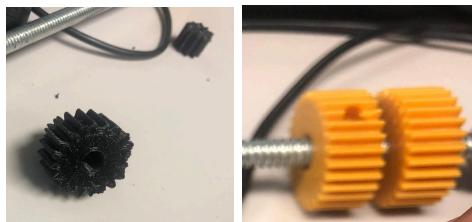
Each of these designs came out to be less than 10cm wide on each side. The two-gear design actually ended up being too small; the extra space was needed for the arm. The three-gear design was slightly better, although space could be optimized. The key problem with both designs was that they ended up using a good amount of space for a 1:1 gear ratio. In my research, I was able to find a website called Gear Generator 2, which helped us learn about gear ratios even further.



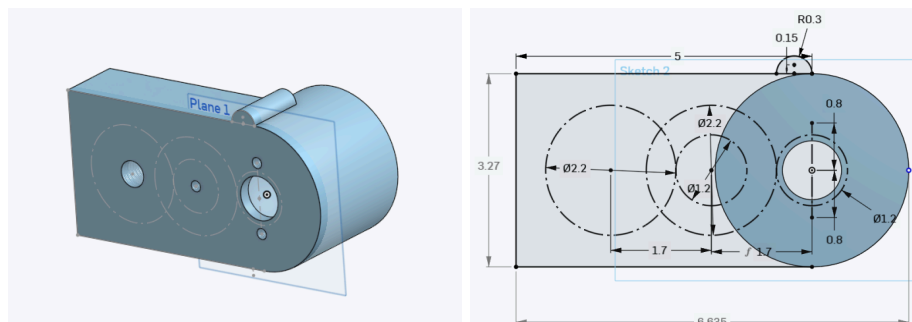
Using the website, I found that to even achieve a 4:1 gear ratio using regular spur gears would take far too much space. Therefore, it would be best to use compound gears. Compound gears often have smaller gears stacked on top of a bigger gear. These types of gears are commonly used in gearboxes and can provide higher torque in a smaller area. Using this knowledge, I was able to achieve a 4:1 gear ratio in less space.



With the help of a website called [STLGears.com](https://www.stlgears.com/), I was able to prototype gears with a 3D printer. Through this, I was able to print out regular spur gears, but designs with overhang could not be printed correctly.



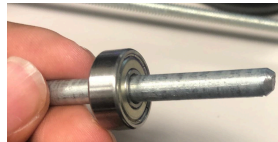
I was able to come up with a design for the gearbox that takes these new gears into account.



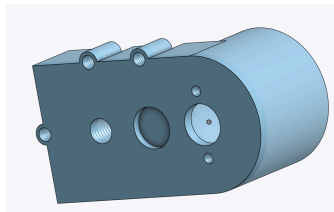
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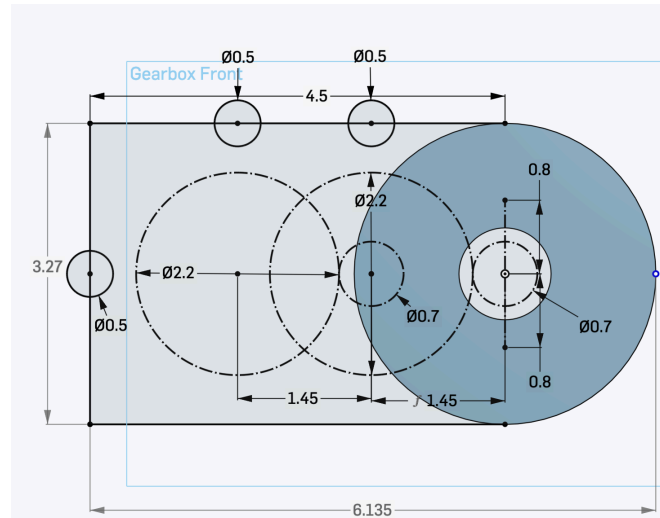
Actuator Gearbox Design

Today, I was able to finalize the gears the actuator will use. The driver gear will be a 10-tooth gear with a **module** of 1. The module of a gear is the ratio of the gear's **pitch circle diameter**, which is its diameter without the teeth in millimeters, to the number of teeth the gear has. All gears that are meshed together must have the same module. All gears will have a module of 1 to ensure precise movement and conserve space. The middle gear will be a compound gear with a 20-tooth bottom gear and a 10-tooth top gear. The top gear will be meshed with another 20-tooth gear that will be attached to the M7 bolt to be rotated. This gearbox has an **8:1** gear ratio. As of this time, the driver gear has been permanently attached to the motor.



With the exception of the driver gear, all other gears require an **axle**, which is typically a rod that provides rotary motion and stability for the gears. Each axle will also require at least one, but generally two bearings, which are machine elements that hold the axle in place and reduce friction during rotation. The bearings I have on hand are plain bearings, which use sliding surfaces with lubricant to reduce friction.





The gearbox has also undergone revisions to reflect these changes. In the middle, there is now space for a bearing with a 6 cm inner diameter (ID). However, this hole is not deep enough, and making it deeper would conflict with the housing of the motor. Depending on the orientation of the gears, this may not be a large issue, but it should be considered. Additionally, there are now holes surrounding the gearbox. These holes are meant for **M4** nuts and bolts that will secure both parts of the gearbox.

Initial Gear Prototyping

Unfortunately, through further prototyping, I found that the gearbox's design would not fit our needs.



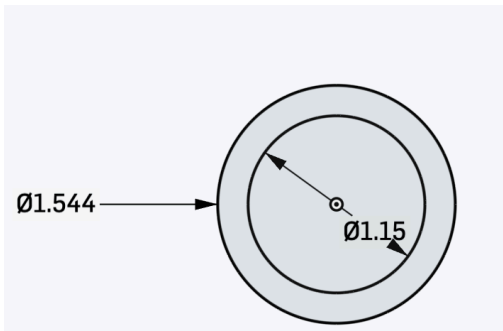
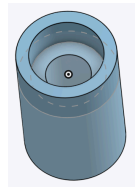
If the gearbox were to utilize only three gears, then there would be no space for the compound gear to mesh with the rod correctly. One solution would be to add a fourth gear that results in the same ratio. This will end up taking some space, but the main problem is that this will require another axle, which also requires another bearing. In short, this would increase the complexity of the design further than necessary. The other solution is to use a bigger gear. Additionally, the compound gear also needs to be redesigned so that the top gear is slightly thicker than the bottom for easier meshing.

Actuator Arm Progress

Progress has also been made on the actuator's arm. The arm will be made of a smaller PVC pipe to save time on fabrication.



Linear actuators of this kind require a **driver nut**. This is the nut that will be used to hold down and displace the arm when the M7 bolt is being rotated by the motor. The idea was to use a cork with a hole to fit the nut in, then place the cork in the middle of the arm (to distribute the load). However, it is important that the hole for the nut is in the middle. Currently, I do not possess the tools to make holes with the required precision, so I will be using a 3D printed design to compensate.



This design was made with the diameters of both the PVC arm and the nut in mind. This design will rely on **friction** to fit both the nut and the cork, then use epoxy to secure them in place.

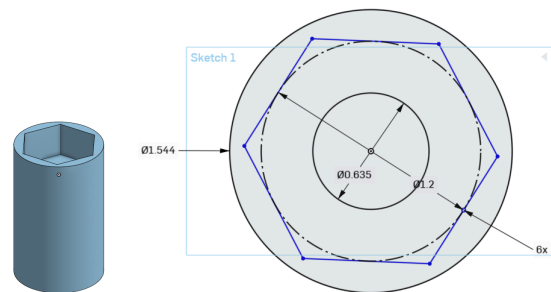
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Name Change

As of today, the float will be named the **Tiger Tower I**. This was not my choice.

Linear Actuator Revisions

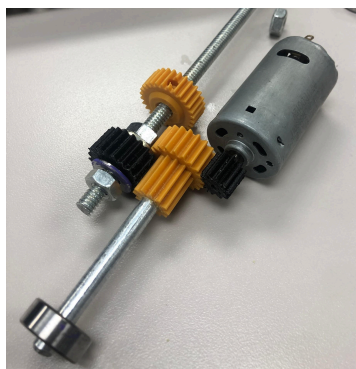
Driver Nut Holder



The holder was redesigned to have a hexagonal shape rather than a circular shape. Instead of relying on friction to fit the nut, it will instead be glued with epoxy. Additionally, the diameter of the holder was slightly increased for better friction. Currently, this is the 5th iteration of this holder, and future iterations will likely have space for nuts on both sides for better load management. Additionally, the driver nut may be placed near the gearbox instead of the middle for better space, but this may affect the way it manages load.

Compound Gear and Bolt Gear

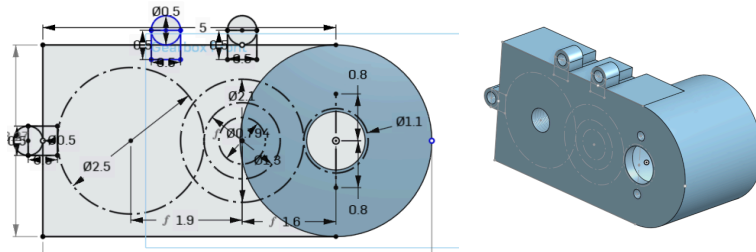
Both the compound gear and bolt gear have been redesigned.



As mentioned on [March 26th](#), the compound gear had various issues that prevented it from being used in the gearbox. To alleviate this, the smaller gear was lengthened for easier meshing, and the diameter has been adjusted to fit the axle.

Additionally, the bolt gear was given an increase in diameter by giving it more teeth, allowing it to maintain its module and gear ratio while increasing the distance between the compound gear and the bolt.

Gearbox Redesign



Each location of the hole has been readjusted to better fit the sizes of the inner diameters of each gear. The reference (construction) circles intersect each other at one or two points so that the gears will mesh correctly.

Remaining Tasks for the Actuator



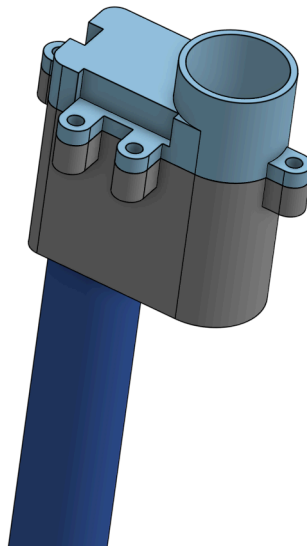
The bolt (which is now identified as a **¼ inch bolt**) is an axle and still requires its own bearing to ensure smooth motion. However, most bearings do not accommodate bolts. Depicted in this picture is an [R4AZZ bearing](#) placed on this bolt. It may seem like the bearing fits perfectly, but there is too much **clearance** (or a difference in space) for the bearing to work properly. There are two ways to get around this.

The first method is to 3D print a spacer to attach to the bolt so that it can be fit into a [608-2RS bearing](#). This, however, will require a great amount of precision to fit properly. It should be noted that **when using friction to fit parts into a space, the perimeter of the space should not be exactly the same as the part.** For example, if there is a 6 cm diameter circular component, it will be nearly impossible to fit it into a 6 cm diameter hole. Instead, the hole should be 6.05 cm to allow for tolerance. **Tolerance** is a deviation from the intended dimension to account for the fact that absolute precision is not guaranteed. A small tolerance can be used for parts that can be hammered in, like the axle bearing, and a (relatively) larger tolerance can be used for parts that can be placed by hand, like the driver nut holder.

The second method is to use a ¼ in. nut and use it as a spacer for an even bigger washer. This will require slightly more space while allowing us more leniency in achieving a high level of precision.

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Linear Actuator



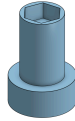
Now that the linear actuator is nearing completion (save for a few key issues), it is time to assemble the other components and prepare the chassis for the assembly of the vertical profiling float.

Full Design of the Gearbox

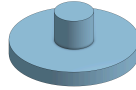
Above is the full design of the gearbox. The motor housing remains unchanged, but the shell itself has been given a **bevel** on the sides to reduce material and increase strength. The orientation of the gearbox has been adjusted so that it will face down, and not up, as seen in previous designs. Finally, the gearbox will be secured with **4** M4 bolts, instead of 3. This is to evenly distribute the load.

Final Arm Revisions

The arm has undergone minor changes since the finalization of the gearbox design.



The driver nut holder was redesigned to have a wider base below the uniform cylinder. This is to cause friction between the pipe that will contain the arm, which is necessary to ensure that the arm itself cannot rotate.

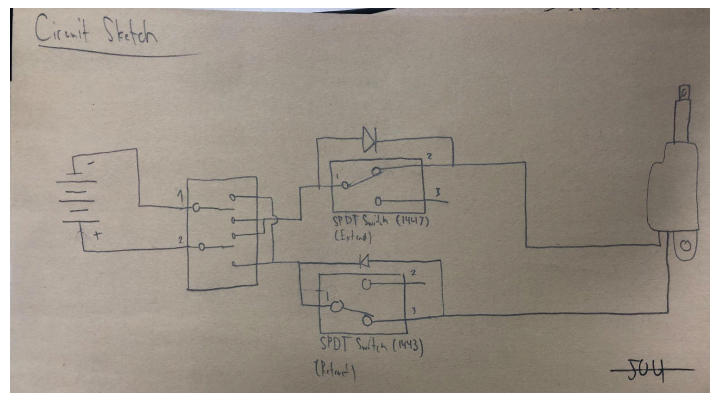


The arm will also have a 2-inch diameter piston to help better distribute the load.

Linear Actuator Circuitry

The linear actuator will require some additional circuitry. One of the main reasons a linear actuator can fail is due to overextension. If there is no limit to how far the arm can extend or retract, the motor can fail, or the arm can be damaged. Such a scenario would be devastating to our float's movement, so it would be best to limit how far the arm can extend.

The first idea was to use diodes to limit the current to one direction so that the arm can only be reversed once it hits a limit.



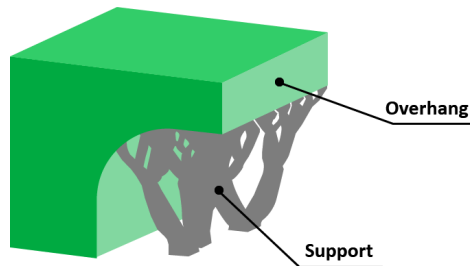
However, the motor will likely be drawing 5 amps of current at maximum, and there will already be a fuse to stop such a thing. What's more dangerous is that the diodes available to us can only handle 1 amp of current, and there is no way to limit it without limiting the motor's power. Therefore, a solution with the microcontroller will need to be developed so that a much smaller current can be used in this way. In any case, limiter switches will need to be collected, and the arm will need a way to toggle them.

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Gearbox Assembly

Gearbox Printing

Using Onshape, I exported the STL files and printed them out. However, there were some key adjustments that made this possible. Namely, I needed to enable **support** on my prints. This is because both parts of the gearbox contain **overhang**. As the name suggests, these sections of the model hang in the air. In other words, these parts are elevated and contain no support on their own. Without some structural support below, the filament will simply fall.



To alleviate this, most 3D printing studios, including Bambu Lab, allow for adding support to overhangs. I specifically used the tree support to save filament and allow for easier removal.



Additionally, I used a **gyroid** infill pattern instead of a rectilinear pattern. In my research, we found that using a gyroid infill pattern could provide greater strength with less material, at the expense of density.

Spacer Solution



As mentioned before, the bolt would require a spacer to fit snugly in the R4AZZ bearing. Additionally, the holes made in the gearbox to fit the bearings were off by a small amount. In retrospect, it would have been more productive to print **test holes** to find the correct diameter needed. Using a digital caliper to find exact measurements was a step in the right direction, but the diameter would need to be adjusted by a small amount to provide sufficient clearance.

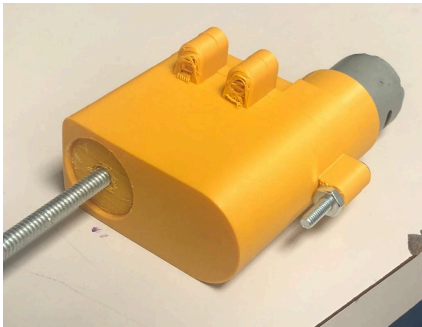
In the meantime, using tape proved to be a fast solution for this problem, as it fills the roles of the spacer while also being removable, if needed. This may become a permanent solution.

Arm Assembly



Both the piston and driver nut have been printed and attached to the PVC pipe. The piston was originally intended to have a cone shape to distribute force better, but due to space constraints, we gave it a 2-inch diameter disk shape instead.

Assembly Process



A small indent was made where the bolt juts out in order to fit the additional PVC pipe or the arm container. However, there were some issues with the diameters of the bearing holes, and so I would need to adjust them again. Additionally, the extrudes for the bolts don't secure the bottom of the linear actuator; they are only able to secure the top and sides. Therefore, I need to adjust the orientation of the holes.

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Electronics Assembly

Because the linear actuator is nearing completion, I thought it would be best to prepare the electronics compartment again. Judging from what I learned 2 months ago, it was best to use a pre-existing container for the compartment.

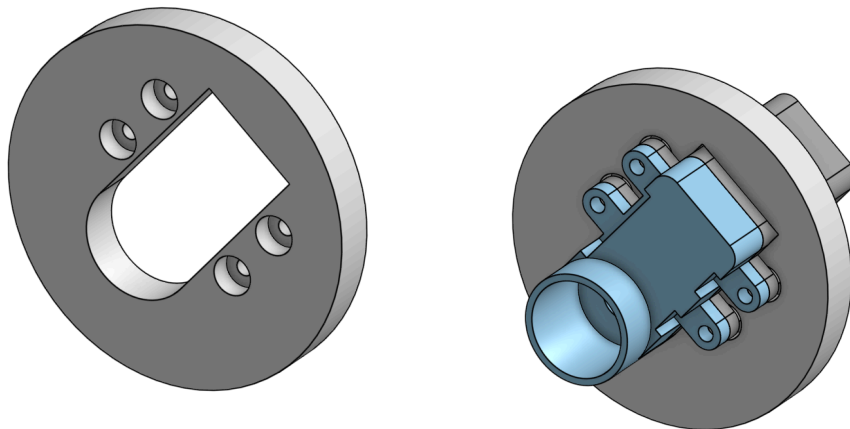
However, most containers I found did not serve much use with my particular design. Instead, I decided to **use the float as its own container** while placing a silicon seal for a platform at the bottom and using the float's o-ring cap as the top seal.

Within this compartment would be 7 components: the batteries, the fuse, the motor controller, the power switch, the pressure sensor, and the microcontroller. The platform will also contain a large hole to fit wires for the limit switches.

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Linear Actuator Platform

While the actuator is coming together, the linear actuator still needs to be affixed to the float somehow. To solve this problem, I began designing a platform that could be used to hold the linear actuator.



The idea was that I could use the cylindrical nature of the float to my advantage by creating a disk that would fit its inner diameter while adding the necessary holes to fit the actuator's gearbox. I would then have 4 holes that the M4 bolts can be used with to secure the actuator itself. However, this disk would have to be thick to be more stable, so I

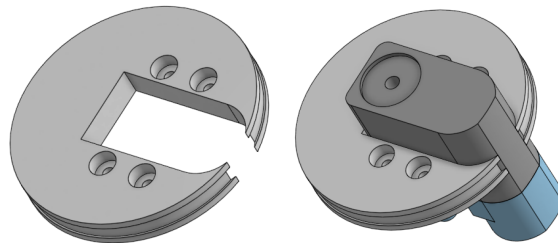
made the holes **counterbore** holes so that I could retain the disk's thickness while also using up less of the bolt's length.

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Platform Redesign

Printing the platform went smoothly. I also decided to use a gyroid pattern here to increase strength, since this part is handling the linear actuator. While assembling the actuator with the platform went as expected, I found that the actuator's arm could not be aligned at the middle of the actuator due to how the disk was designed. It is important that the arm is in the middle to distribute the load properly.

To solve this problem, I simply shifted the hole where the actuator was supposed to be to the right. However, this created a new problem: the gearbox would now escape the platform's diameter. To alleviate this, I simply removed the section of the gearbox that was outside the diameter. This ended up being a quick solution with the added side effect of allowing the platform to be friction-fit. Originally, I had indents on the side of the platform that would allow me to increase the effective diameter of the platform with a rubber band if a friction-fit was not possible naturally. This was meant to be temporary; it was simply a way to finalize the platform's placement before gluing it to the inner chassis.



Electronics + Interface Development

Today, I was able to take my Raspberry Pi Pico W home to program the interface. While I was not able to secure any alternative modules or microcontrollers, I believe that this alone will be sufficient for our operations.

A key part of my choice to use the Raspberry Pi Pico W was its innate capability to use Wi-Fi. Using this, I am able to host a server with the Pico, which allows me to send it commands through a website interface.

The Raspberry Pi Pico also possesses a way to convert the 4.8V input into a 5V input, which can be used to power the L298N motor controller I've been provided. It can also be used to power and support an SD card adapter. Having the data for every profile can streamline testing, but due to time constraints, it will likely have to be removed. Like most

microcontrollers, the Pico also has **analog** pins, which will be used to read the input from the pressure sensor. **These pins can go from 0V to 3.3V, and I can write code to interpret the voltage as usable data, like the current depth of the float.** Two digital pins will be dedicated to the limit switches, while two more will be used to control the motors using the controller.

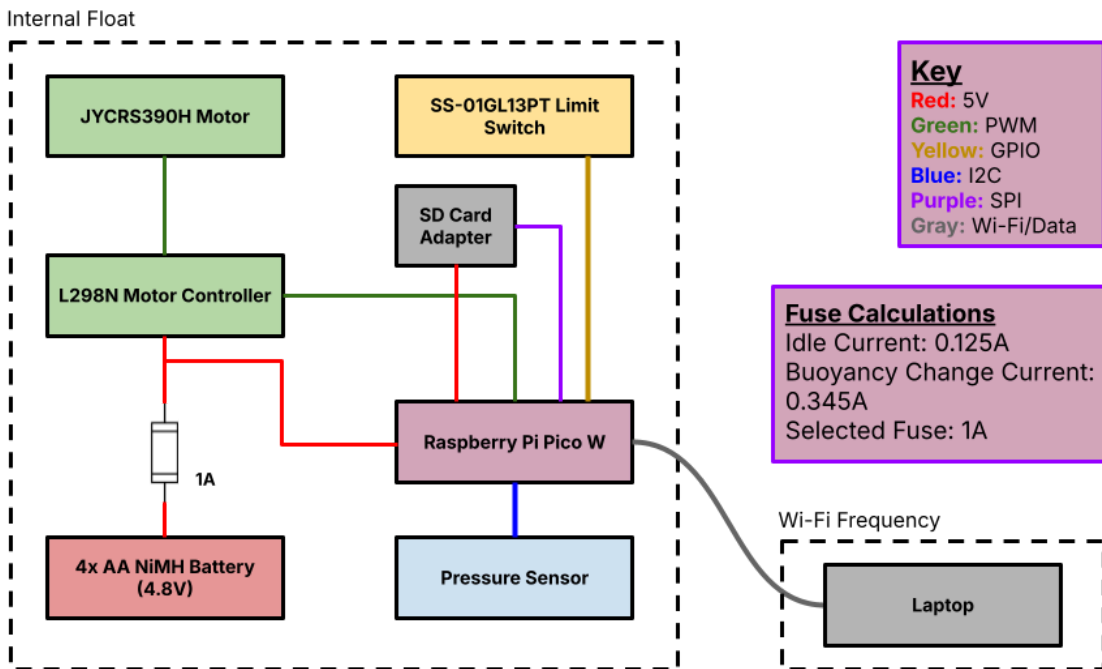


Figure 4. Tiger Tower I SID

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Conclusion of Construction

Despite the progress made with the linear actuator and its interface, unfortunately, I will not have time to complete this float in time for the Houston MATE 2026 competition. There are problems with our ROV, Hammerhead I, that must be addressed quickly before Friday. Additionally, with AP exams rolling around in addition to being shorthanded, I will have to attempt to fix some of Hammerhead I's problems while also practicing piloting. However, this competition is one that exemplifies iteration and posterity. Because of that, it would be best to leave the float as it is now without making any permanent changes to the chassis in the hopes that this logbook can be used to finish or adapt the project by my juniors.